Vector Fields and Line Integrals

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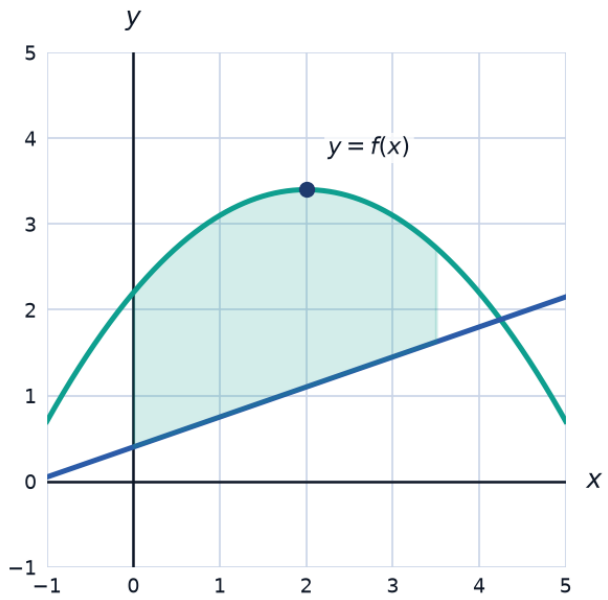
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Vector fields and divergence

For the vector field $\mathbf{F}(x, y) = \langle x^2 y,\ xy^2 \rangle$, find the divergence $\nabla \cdot \mathbf{F}$.

2 [Foundational] • Target: Vector fields and divergence

For $\mathbf{F}(x, y, z) = \langle y,\ -x, z^2 \rangle$, find the curl $\nabla \times \mathbf{F}$.



3 [Foundational] • Target: Line integrals

Evaluate the line integral $\int_C (x + y)\, ds$ where C is the line segment from $(0, 0)$ to $(3, 4)$.

4 [Foundational] • Target: Line integrals

Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle y, x \rangle$ along $\mathbf{r}(t) = \langle t, t^2 \rangle$, $0 \le t \le 1$.

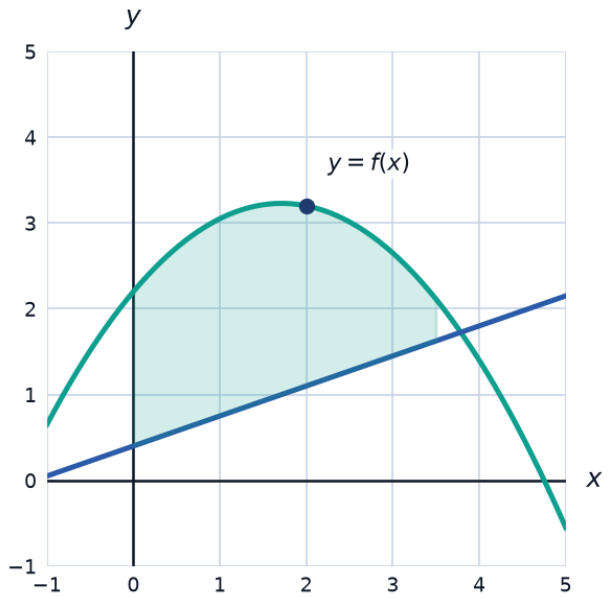
5 [Foundational] • Target: Conservative vector fields and potential functions

Determine whether $\mathbf{F}(x, y) = \langle 2xy, x^2 + 3y^2 \rangle$ is conservative, using the test $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$.

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6 [Foundational] • Target: Conservative vector fields and potential functions

Find a potential function $f(x, y)$ such that $\nabla f = \mathbf{F}$ for the field above.



7 [Foundational] • Target: The Fundamental Theorem for line integrals

Using the potential function $f(x, y) = x^2y + y^3$ found above, evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(0, 0)$ to $(1, 2)$, using the Fundamental Theorem for line integrals.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Green's Theorem

Use Green's Theorem, $\oint_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$, to evaluate $\oint_C xy\,dx + x^2\,dy$ where C is the boundary of the square $0 \le x \le 1$, $0 \le y \le 1$.

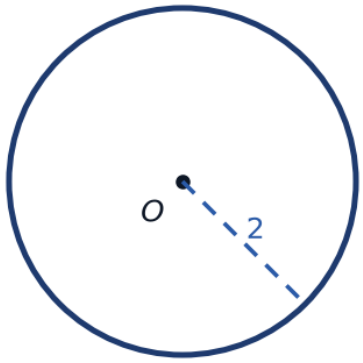
9 [Intermediate] • Target: Green's Theorem

Use Green's Theorem to find the area of the region enclosed by a curve C, using the formula $A = \frac{1}{2}\oint_C (x\,dy - y\,dx)$, for the circle $x = 2\cos t$, $y = 2\sin t$, $0 \le t \le 2\pi$.

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10 [Intermediate] • Target: Green's Theorem

Use Green's Theorem to evaluate $\oint_C (3y)\,dx + (5x)\,dy$ where C is the circle $x^2 + y^2 = 4$, traversed counterclockwise.



11 [Intermediate] • Target: Work done by a force field

A force field $\mathbf{F}(x, y) = \langle y, x \rangle$ acts on an object moving along the straight path from $(0, 0)$ to $(2, 0)$ then from $(2, 0)$ to $(2, 3)$. Find the total work done, $W = \int_C \mathbf{F} \cdot d\mathbf{r}$, by evaluating each leg separately.

12 [Intermediate] • Target: Work done by a force field

Confirm the work found above using the potential function $f(x, y) = xy$ (since $\mathbf{F} = \langle y, x \rangle = \nabla f$) and the Fundamental Theorem for line integrals.

13 [Intermediate] • Target: Testing for conservative fields

Determine whether $\mathbf{F}(x, y) = \langle y^2, 2xy + 1 \rangle$ is conservative, and if so, find a potential function.

14 [Intermediate] • Target: Testing for conservative fields

Determine whether $\mathbf{F}(x, y) = \langle 3x^2 y, x^3 + 2y \rangle$ is conservative, and find a potential function if so.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

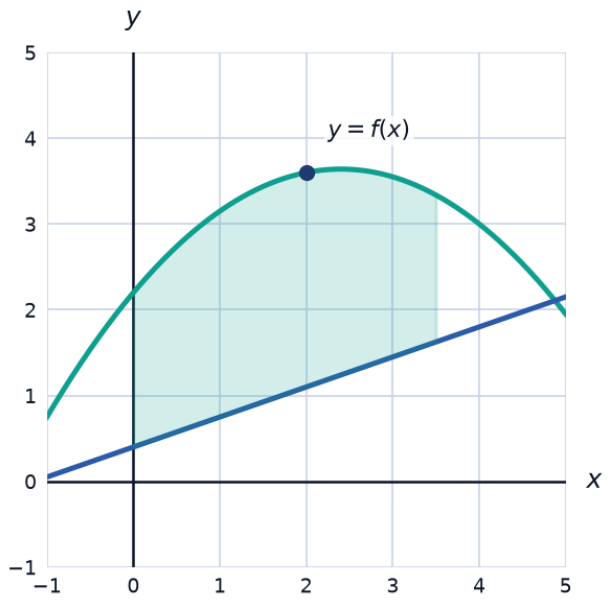
15 [Challenging] • Target: Applying Green's Theorem to circulation • Cross-Lesson: 9.01 Separable Differential Equations

Use Green's Theorem to evaluate the circulation $\oint_C (-y)\,dx + x\,dy$ around the boundary of the rectangle $0 \le x \le 3$, $0 \le y \le 2$, traversed counterclockwise.

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16 [Challenging] • Target: Line integrals over piecewise paths • Cross-Lesson: 1.01 Precalculus Review for Calculus

Evaluate $\int_C y\,dx + x\,dy$ where C consists of the segment from $(0, 0)$ to $(1, 0)$ followed by the segment from $(1, 0)$ to $(1, 1)$.



17 [Challenging] • Target: Line integrals over piecewise paths • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Confirm the result above using the potential function $f(x, y) = xy$ (since $\langle y, x\rangle = \nabla f$) and the endpoints $(0, 0)$ and $(1, 1)$.

18 [Challenging] • Target: Flux and Green's Theorem (normal form) • Cross-Lesson: 11.01 Sequences and Series

Using the divergence form of Green's Theorem, $\oint_C \mathbf{F}\cdot\mathbf{n}\,ds = \iint_D \nabla\cdot\mathbf{F}\,dA$, find the outward flux of $\mathbf{F}(x, y) = \langle x, y\rangle$ across the boundary of the disk $x^2 + y^2 \le 4$.

19 [Challenging] • Target: A capstone conservative-field application • Cross-Lesson: 12.01 Convergence Tests

A particle moves through a force field $\mathbf{F}(x, y) = \langle 2xy^3, 3x^2y^2 + 4\rangle$ from the point $(0, 0)$ to $(2, 1)$ along any path. First verify that $\mathbf{F}$ is conservative, then find a potential function $f(x, y)$, and finally use it to compute the total work done by the field on the particle.

20 [Challenging] • Target: A capstone conservative-field application • Cross-Lesson: 12.02 Power Series and Taylor Series

For the vector field $\mathbf{F}(x, y, z) = \langle yz, xz, xy + 2z\rangle$, complete each step:

a Find the divergence $\nabla\cdot\mathbf{F}$.

b Find the curl $\nabla\times\mathbf{F}$.

c Based on the curl, determine whether $\mathbf{F}$ could be conservative in 3D.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Vector Fields, Divergence, and Curl

Compute the divergence $\nabla \cdot \mathbf{F}$ for $\mathbf{F} = \langle x^2 y, 3yz, -2xz \rangle$.

a. $2xy + 3z - 2x$

b. $2xy + 3y - 2z$

c. $x^2 + 3z - 2x$

d. 0

22 [Foundational] • Target: Vector Fields, Divergence, and Curl

Compute the 2D curl of $\mathbf{F} = \langle -y, x \rangle$.

a. 2

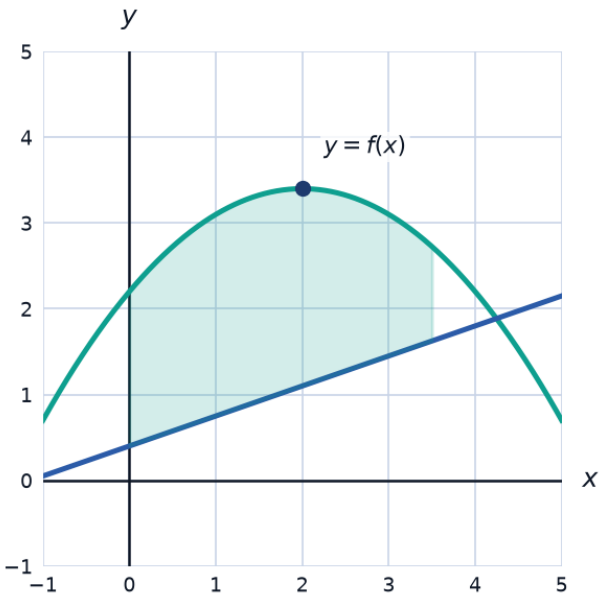
b. 0

c. 1

d. -2

23 [Foundational] • Target: Vector Fields, Divergence, and Curl

If a vector field has $\nabla \times \mathbf{F} = \mathbf{0}$ throughout a simply connected domain, what is it called?



a. Conservative (or irrotational)

b. Incompressible

c. Solenoidal

d. Harmonic

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24 [Foundational] • Target: Line Integrals with Respect to Arc Length

State the formula for $\int_C f(x,y)\,ds$ along a curve parameterized by $\mathbf{r}(t)$ for $a \le t \le b$.

a. $\int_a^b f(x(t),y(t))\sqrt{[x'(t)]^2+[y'(t)]^2}\,dt$

b. $\int_a^b f(x(t),y(t))\,dt$

c. $\int_a^b f(x(t),y(t))[x'(t)+y'(t)]\,dt$

d. $\int_a^b f'(t)\,dt$

25 [Foundational] • Target: Line Integrals with Respect to Arc Length

Evaluate $\int_C (x+y)\,ds$ where C is the line segment from $(0,0)$ to $(3,4)$.

a. $\frac{35}{2}$

b. 35

c. 7

d. 12

26 [Foundational] • Target: Line Integrals with Respect to Arc Length

Find the mass of a wire bent in the shape of the semicircle $\mathbf{r}(t) = \langle 2\cos t, 2\sin t\rangle$ $(0 \le t \le \pi)$ with density $\rho(x,y) = y$.

a. 4

b. 8

c. 2π

d. 4π

27 [Foundational] • Target: Line Integrals of Vector Fields (Work Integrals)

Evaluate $\int_C \mathbf{F}\cdot d\mathbf{r}$ for $\mathbf{F} = \langle y, -x\rangle$ along the circle $\mathbf{r}(t) = \langle \cos t, \sin t\rangle$ for $0 \le t \le 2\pi$.

a. -2π

b. 2π

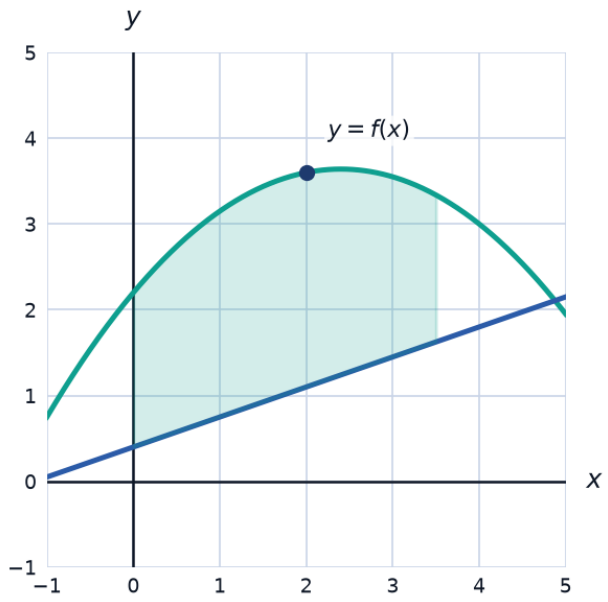
c. 0

d. π

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28 [Foundational] • Target: Line Integrals of Vector Fields (Work Integrals)

Find the work done by the constant force $\mathbf{F} = \langle 3, 5 \rangle$ moving an object along the straight line from $(1, 1)$ to $(4, 5)$.



a. 29

b. 24

c. 32

d. 15

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Line Integrals of Vector Fields (Work Integrals)

Evaluate $\int_C (x\,dx + y\,dy)$ along the parabolic arc $y = x^2$ from $(0, 0)$ to $(1, 1)$.

a. 1

b. 1/2

c. 2/3

d. 0

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30 [Intermediate] • Target: Fundamental Theorem of Line Integrals and Green's Theorem

State the Fundamental Theorem of Line Integrals for a conservative field $\mathbf{F} = \nabla f$.

a. $\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$

b. $\int_C \nabla f \cdot d\mathbf{r} = 0$ for all curves

c. $\int_C \nabla f \cdot d\mathbf{r} = ||\nabla f||$

d. $\int_C \nabla f \cdot d\mathbf{r} = f'(\mathbf{r}(b))$

31 [Intermediate] • Target: Fundamental Theorem of Line Integrals and Green's Theorem

State Green's Theorem relating a line integral along a positively oriented boundary ∂D to a double integral over D.

a. $\oint_C (P\,dx + Q\,dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$

b. $\oint_C (P\,dx + Q\,dy) = \iint_D (P + Q)\,dA$

c. $\oint_C (P\,dx + Q\,dy) = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}\right) dA$

d. $\oint_C (P\,dx + Q\,dy) = 0$

32 [Intermediate] • Target: Fundamental Theorem of Line Integrals and Green's Theorem

Use Green's Theorem to evaluate $\oint_C (3y\,dx + 5x\,dy)$ where C is the unit circle $x^2 + y^2 = 1$ oriented counterclockwise.

a. 2π

b. 8π

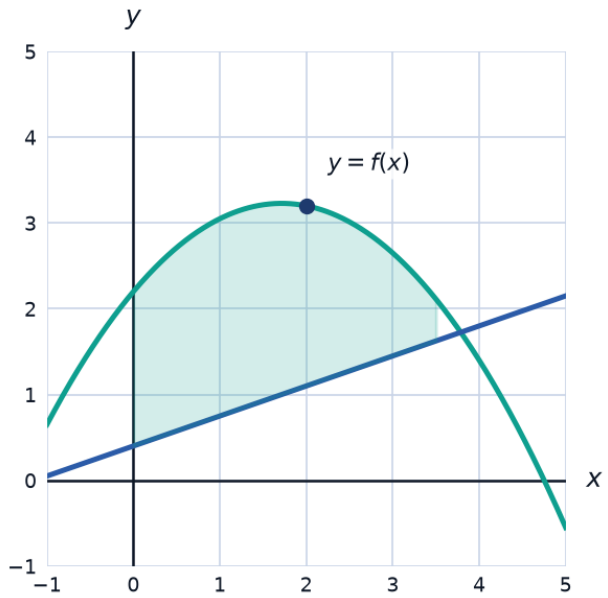
c. 0

d. π

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33 [Intermediate] • Target: Line Integrals with Respect to Arc Length

Find the work done by $\mathbf{F} = (x^2, y^2, z^2)$ along line segment from $(0, 0, 0)$ to $(1, 1, 1)$:



a. 1

b. $\frac{2}{3}$

c. 3

d. $\frac{1}{2}$

34 [Intermediate] • Target: Line Integrals of Vector Fields (Work Integrals)

Divergence theorem: $\oiint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V (abla \cdot \mathbf{F})\, dV$. For $\mathbf{F} = \mathbf{r} = (x, y, z)$, $abla \cdot \mathbf{F} = 3$: flux through unit sphere =

a. 3

b. $3V = 4\pi$

c. 4π

d. 12π

35 [Intermediate] • Target: Fundamental Theorem of Line Integrals and Green's Theorem

Work done by $\mathbf{F} = (y, x, 0)$ along line from $(0, 0, 0)$ to $(1, 1, 0)$:

a. 0

b. 2

c. 1

d. 1/2

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36 [Intermediate] • Target: Vector fields and divergence

The divergence theorem applied to $\mathbf{F} = (x^3, y^3, z^3)$ over unit ball gives $\oiint_S \mathbf{F} \cdot \hat{n}\, dS = \iiint_V 3(x^2 + y^2 + z^2) dV =$

a. $4\pi/3$

b. 3π

c. $12\pi/5$

d. π

37 [Intermediate] • Target: Line integrals

Green's theorem in the form $\iint_D (\partial Q/\partial x - \partial P/\partial y) dA = \oint_C P dx + Q dy$. For $P = 0, Q = x$: $\oint_C x\, dy =$

a. $-\iint_D dA$

b. 0

c. $\iint_D dA = \text{Area}(D)$

d. $\oint_C y\, dx$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Conservative vector fields and potential functions • Cross-Lesson: 11.01 Sequences and Series

Green's theorem: $\oint_C (P\, dx + Q\, dy) = \iint_D (Q_x - P_y)\, dA$. For $P = -y, Q = x$:

a. area $= \frac{1}{2}\oint (x\, dy - y\, dx)$

b. area $= \iint dA = 2$ times the line integral

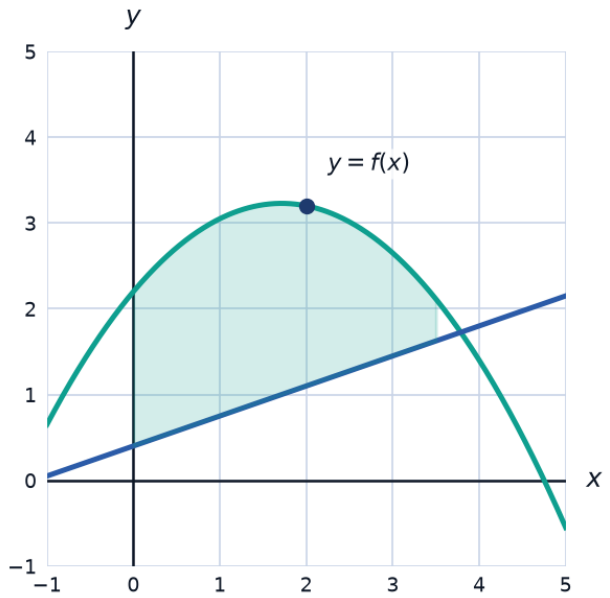
c. perimeter $= \oint ds$

d. zero

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39 [Challenging] • Target: The Fundamental Theorem for line integrals • Cross-Lesson: 12.01 Convergence Tests

A conservative vector field $\mathbf{F} = ablaf$ has $abla \times \mathbf{F} =$



a. $\mathbf{F}$

b. f

c. $ablaf$

d. $\mathbf{0}$

40 [Challenging] • Target: Green's Theorem • Cross-Lesson: 12.02 Power Series and Taylor Series

Line integral of scalar field: $\int_C f\,ds = \int_a^b f(\mathbf{r}(t))|\mathbf{r}'(t)|dt$. For $f = 1$, $\int_C 1\,ds =$

a. area

b. volume

c. 0

d. arc length

41 [Challenging] • Target: Work done by a force field • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Normal line to $z = x^2 + y^2$ at $(1, 1, 2)$: direction:

a. $(2, 2, -1)$

b. $(1, 1, 1)$

c. $(-2, -2, 1)$

d. $(1, 0, -2)$

42 [Challenging] • Target: Testing for conservative fields • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Find the flux of $\mathbf{F} = \langle x, y, z \rangle$ through the closed unit sphere using the Divergence Theorem.

a. 2π

b. $\frac{4\pi}{3}$

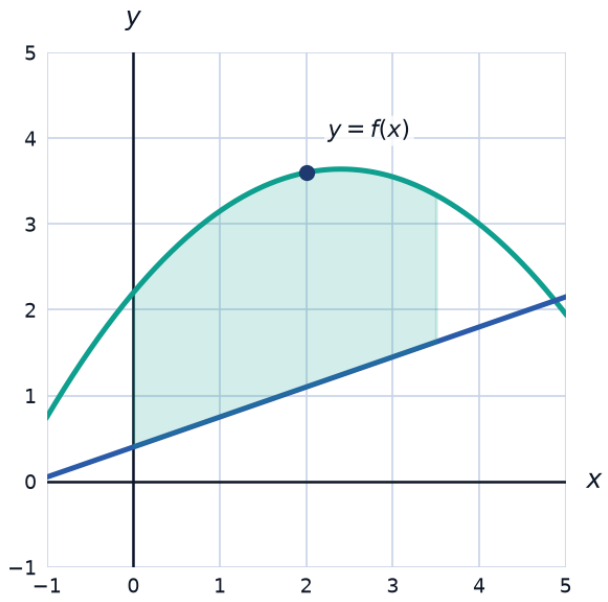
c. 4π

d. $\frac{8\pi}{3}$

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43 [Challenging] • Target: Applying Green's Theorem to circulation • Cross-Lesson: 15.01 Double Integrals

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x^2, y^2, z^2 \rangle$ across the closed cylinder $x^2 + y^2 \le 4,\ 0 \le z \le 2$.



a. 32π

b. 8π

c. -16π

d. 16π

44 [Challenging] • Target: Line integrals over piecewise paths • Cross-Lesson: 2.01 Evaluating Limits

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle xz, 0, z^2 \rangle$ across the closed cylinder $x^2 + y^2 \le 1,\ 0 \le z \le 1$.

a. 3π

b. $\frac{3\pi}{4}$

c. $\frac{3\pi}{2}$

d. $-\frac{3\pi}{2}$

45 [Challenging] • Target: Flux and Green's Theorem (normal form) • Cross-Lesson: 2.02 Continuity

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x, y, z \rangle$ across the closed cylinder $x^2 + y^2 \le 4,\ 0 \le z \le 3$.

a. 36π

b. 72π

c. 18π

d. -36π

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